



10555 - Probing Lyman alpha Escape from a Lyman alpha Emitter Cross at $z=5.7$

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1		NIRSpec IFU Spectroscopy	(5) Group Lya_Cross

ABSTRACT

The escape of Lyman-alpha (Ly α) photons from high-redshift galaxies is key to probing reionization, yet it remains poorly understood and lacks observational constraints. We propose NIRSpec IFU observations of a unique “LAE Cross” at $z=5.7$: four luminous Ly α emitters (LAEs) separated by only 2-5". Our primary goal is to determine precise spectroscopic redshifts for the four components and the central galaxy. If the four components share identical kinematics, then the system is confirmed as a single gravitationally lensed galaxy, and the differing Ly α profiles of the four images will provide an unprecedented probe of the circumgalactic and intergalactic medium on small physical scales, directly constraining Ly α radiative transfer models. Alternatively, if the components show distinct kinematics, then the system is a rare protocluster of four massive LAEs. In this equally compelling scenario, our observations will characterize early galaxy formation in a dense environment.

Crucially, the IFU data will also yield direct measurements of the gas-phase metallicity via the [OIII]4363 line, along with dust content and star formation rates for each component. These physical properties are essential for interpreting the Ly α data in either scenario, as they regulate the

mechanisms of Ly α escape. This program will transform the LAE Cross into a benchmark system for understanding the visibility of galaxies in the early Universe.

OBSERVING DESCRIPTION

We will use the NIRSpec IFU to target four unique $z=5.7$ Lyman-alpha emitters in a custom 4-pointing mosaic. We will observe the rest-frame optical emission lines H-alpha, H-beta, H-gamma, and [OIII]5007,4959,4363 in these objects using the G395M/F290LP filter/grating combination. These emission lines will permit the calculation of systemic redshifts, dust attenuations, spatial dust distributions, star formation rates, kinematics, electron temperatures, gas-phase metallicities, and ages for the targeted ULLAEs. To best optimize time on source and background subtraction, we will use the 4-POINT-DITHER scheme for all targets. For each DITHER position, we will integrate for 61 groups of NRSIRS2RAPID for a total of 1 hour of exposure per pointing. We will perform target acquisition using WATA in the NRSRAPID6/SUB2048/Clear configuration.

Proposal 10555 - Targets - Probing Lyman alpha Escape from a Lyman alpha Emitter Cross at $z=5.7$

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	Lya_A	RA: 22 18 9.0354 (334.5376475d) Dec: +01 07 23.70 (1.12325d) Equinox: J2000 Comments: Category=Galaxy Description=[High-redshift galaxies, Lyman-alpha galaxies]		
(2)	Lya_B	RA: 22 18 9.2035 (334.5383479d) Dec: +01 07 25.21 (1.12367d) Equinox: J2000 Comments: Category=Galaxy Description=[High-redshift galaxies, Lyman-alpha galaxies]		
(3)	Lya_C	RA: 22 18 9.1810 (334.5382542d) Dec: +01 07 28.91 (1.12470d) Equinox: J2000 Comments: Category=Galaxy Description=[High-redshift galaxies, Lyman-alpha galaxies]		
(4)	Lya_D	RA: 22 18 8.9682 (334.5373675d) Dec: +01 07 26.39 (1.12400d) Equinox: J2000 Comments: Category=Galaxy Description=[High-redshift galaxies, Lyman-alpha galaxies]		
(5)	Group Lya_Cross	Comments: Target Selection=[1 Lya_A, 2 Lya_B, 3 Lya_C, 4 Lya_D]		

Proposal 10555 - Observation 1 - Probing Lyman alpha Escape from a Lyman alpha Emitter Cross at z=5.7

Observation	Proposal 10555, Observation 1												Tue May 26 23:00:38 GMT 2026
	Diagnostic Status: Warning												
Diagnostics	Observing Template: NIRSpec IFU Spectroscopy												
	(Visit 1:1) Warning (Form): Data Excess over lower threshold												
Fixed Targets	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
	#	Name			Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
Template	(5) Group Lya_Cross												
	Comments: Target Selection=[1 Lya_A, 2 Lya_B, 3 Lya_C, 4 Lya_D]												
Dithers	TA Method												
	HFF Readout Mode												
Spectral Elements	NONE												
	false												
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points		
	1 4-POINT-DITHER												
Spectral Elements	#	Grating/Filter		Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1 G395M/F290LP NRSIRS2RAPID 61 1 false true NONE 4 4 3618.045												